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Statistical Models

Week 10: Numerical Methods and Markov Chain Monte Carlo 1: the Gibbs sampler

We will start with the "Bayesian CLT", a fast approximation method to find posterior summaries.

Next, we will discuss the general idea of Markov Chain Monte Carlo methods. These methods are based on results for Markov chains – we will shortly review these results. You may know Markov chains with finitely many possible outcomes, that is, the values are integers (finite state space). Here, we use Markov chains with real valued outcomes (continuous state space).

Friday lecture concentrates on the *Gibbs sampler*, a method that allows to generate random samples from the posterior distribution a multidimensional parameter. In essence it consists in updating one element of the multidimensional parameter at a time.

The other very important MCMC method is the *Metropolis-Hastings algorithm*. It enables us to simulate from a distribution that is known up to a constant. If time allows, we will start with the Metropolis algorithm in the end of Friday's lecture.

In order to understand the algorithms, you will implement them yourselves in R. Later in this course you will use software "off the shelf" to run MCMC. You will learn to use JAGS and the R package `rjags` to analyze simple Bayesian models.

Relevant Readings

- Numerical methods: R&G, Section 3.1
- Notes on Markov chains with finite and continuous state space
- Gibbs sampler: Reich & Ghosh, Section 3.2, 3.2.1.
- Why does the Gibbs sampler work? (proof)

Lecture Material

- Link to onenotes on Google drive
- Code for R&G, Figure 3.2
- (evt.) Experiments with AR(1) chains
- Gibbs sampler for bivariate normal distribution
- Listing 3.2 from Reich & Ghosh for experimentation

Important: Install JAGS

In the coming weeks, we will use the R package `rjags` for Bayesian analysis (for Pythonists: use the package `PyJAGS`). `rjags` (or `PyJAGS`) is an interface to the software `JAGS` (Just Another Gibbs Sampler) that you need to install separately on your computer (not from within R). Please install `JAGS` from `sourceforge` (link).

Exercises 10

Problem 10.1: MAP and numerical integration

Reich & Ghosh, Chapter 3, Exercise 2

Assume that $Y \mid \mu \overset{\text{indep.}}{\sim} \text{Normal}(\mu, \sigma_i^2)$, for $i = 1, \ldots, n$, with σ_i known and improper prior distribution $\pi(\mu) = 1$ for all μ.

a. Give a formula for the MAP estimator for μ.

b. We observe $n = 3, Y_1 = 12, Y_2 = 10, Y_3 = 22$, and know that $\sigma_1 = \sigma_2 = 3$ and $\sigma_3 = 10$. Compute the MAP estimate of μ.

c. Use numerical integration to compute the posterior mean of μ.

d. Plot the posterior distribution of μ and indicate the MAP and the posterior mean estimates on the plot.

Hints

Parts a) and b) can be solved without R; you might want to compare your analytical result for the MAPin b) with numerical results by R function `optim`.
For parts c) and d), find ideas in listing 3.1, R&G page 72. You can use function `adaptIntegrate` from package `cubature`, or just work with base R function `integrate` here. You can download the R code here.
Both functions calculate the integral of a function that can take a vector of all arguments and would return a vector of all function values. If your own function only returns a single number, you will get weird results. The remedy is then to create a new version of your function by applying R function `Vectorize`.

Problem 10.2: Bayesian CLT approximation

Reich & Ghosh, Chapter 3, Exercise 3.

Assume $Y_i \mid \lambda \overset{\text{indep}}{\sim} \text{Poisson}(N_i\lambda)$ for $i = 1, \ldots, n$.

a. Identify a conjugate prior for λ and derive the posterior that follows from this prior.

b. Using the prior $\lambda \sim \text{Uniform}(0, 20)$, derive the MAP estimator of λ.

c. Using the prior $\lambda \sim \text{Uniform}(0, 20)$, plot the posterior on a grid of λ assuming $n = 2, N_1 = 50, N_2 = 100, Y_1 = 12$, and $Y_2 = 25$ and demonstrate that the MAP estimate is indeed the maximizer (draw a vertical line).

d. Use the Bayesian CLT to approximate the posterior of λ under the setting of (c), plot the approximate posterior, and compare the plot with the plot from (c).

Hint

For d), find code for Figure 3.2 (R&G) on the course supplement page

Try at least one of the following two problems about Gibbs samples; we will get back to them later:

Problem 10.3: Inference for the variance in a hierarchical normal model

Reich & Ghosh, Chapter 3, Exercise 4 (a) – (d).

Consider the model $Y_i \mid \sigma_i^2 \overset{\text{indep}}{\sim} \text{Normal}(0, \sigma_i^2)$ for $i = 1 \ldots n$, where $\sigma_i^2 \mid b \sim \text{InvGamma}(a, b)$ and $b \sim \text{Gamma}(1, 1)$.

a. Derive the full conditional posterior distributions for σ_1^2 and b.

b. Write pseudocode for Gibbs sampling, i.e., describe in detail each step of the Gibbs sampling algorithm.

c. Write your own Gibbs sampling code (not in JAGS) and plot the marginal posterior density for each parameter. Assume $n = 10$, $a = 10$ and $Y_i = i$ for $i = 1, \ldots 10$.

d. Repeat the analysis with $a = 1$ and comment on the convergence of the MCMC chain.

% You can skip JAGS question e. e. Implement this model in (c) using JAGS and compare the results with the results in (c).

Problem 10.4: Gibbs sampling for comparison of two groups in normal model

Reich & Ghosh, Chapter 3, Exercise 5 (a) – (c).

Consider the model $Y_i \mid \mu.\sigma^2 \sim \text{Normal}(\mu, \sigma^2)$ for $i = 1, \ldots, n$ and $Y_i \mid \mu, \sigma^2 \sim \text{Normal}(\mu + \delta, \sigma^2)$ for $i = m + 1, \ldots, n + m$, where $\mu, \delta \sim \text{Normal}(0, 100^2)$ and $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$.

a. Give an example of a real experiment for which this would be an appropriate model.

b. Derive the full conditional posterior distributions for μ, δ, and σ^2.

c. Simulate a dataset from this model with $n = m = 50$, $\mu = 10, \delta = 14$, and $\sigma = 2$. Write your own Gibbs sampling code (not in JAGS) to fit the model above to the simulated data and plot the marginal posterior density for each parameter. Are you able to recover the true values reasonably well? Try to find out if your result in (c) varies a lot when you simulate again.

%You can skip JAGS question d. (d) Implement this model using JAGS and compare the results with the results in (c).

The last problem expands on the AR(1) example for a Markov chain and illustrates that a Markov chain of length n does not have the same properties as an i.i.d. sequence of length n.

Problem 10.5: Interesting and relevant properties of the AR(1) process

In the lectures, we discuss(ed) the autoregressive process AR(1) as an example for a Markov chain (see the interactive experiment)
Consider the AR(1) chain $X_0, X_1, \ldots$ given by

$$X_n = \phi X_{n-1} + \lambda + \tau Z_n, \quad n = 1, 2, \ldots,$$

where

$$Z_1, Z_2, \ldots \overset{\text{i.i.d.}}{\sim} \text{N}(0, 1), \; -1 < \phi < 1, \; \lambda \in \mathbb{R}, \; \tau > 0,$$

and Z_n is independent of X_m for all n and $m < n$.

a. Give an argument that the X_n in such a chain must have a normal distribution. Thus, the stationary distribution must be a normal distribution.

   Show that the normal distribution with mean $\mu = \lambda/(1 - \phi)$ and variance $\sigma^2 = \tau^2/(1 - \phi^2)$ is the stationary distribution of the chain.

   Run some simulation(s) to confirm this result.

b. Confirm by calculation or simulation, that, for all $n \in \mathbb{N}$ and $k = 0, 1, \dots, n-1$

$$\text{Cov}(X_n, X_{n-k}) = \phi^k \sigma^2, \quad \text{or, equivalently,} \quad \text{Cor}(X_n, X_{n-k}) = \phi^k.$$

c. Investigate the variance of the sample mean

$$\overline{X}_{(m)} = \frac{1}{m} \sum_{i=1}^{m} X_i$$

from values $X_1, \dots, X_m$ from the AR(1) chain started in a random value X_0 from the stationary distribution, and compare with the variance that you would get when the samples $X_1, \dots, X_m$ were drawn *independently* from the stationary distribution.

Do this by simulating for several values of ϕ, also for some negative ones. What can you conclude?